Unit 3: Environmental Pollution

Topics needs to be covered: Definition Causes, effects, and control measures of: Air pollution; Water pollution; Soil pollution; Marine pollution; Noise pollution; Thermal pollution; nuclear hazards

AIR POLLUTION

Air pollution refers to any physical, chemical or biological change in the air. It is the contamination of air by harmful gases, dust and smoke which affects plants, animals and humans drastically. There are a certain percentage of gases present in the atmosphere. An increase or decrease in the composition of these gases is harmful to survival. This imbalance in the gaseous composition has resulted in an increase in earth's temperature, which is known as global warming**.**

Following are the important causes of air pollution:

Burning of Fossil Fuels: The combustion of fossil fuels emit a large amount of sulphur dioxide. Carbon monoxide released by incomplete combustion of fossil fuels also results in air pollution.

Automobiles: The gases emitted from vehicles such as jeeps, trucks, cars, buses, etc. pollute the environment. These are the major sources of greenhouse gases and also result in diseases among individuals.

Agricultural Activities: Ammonia is one of the most hazardous gases emitted during agricultural activities. The insecticides, pesticides and fertilizers emit harmful chemicals in the atmosphere and contaminate it.

Factories and Industries: Factories and industries are the main source of carbon monoxide, organic compounds, hydrocarbons and chemicals. These are released into the air, degrading its quality.

Mining Activities: In the mining process, the minerals below the earth are extracted using large pieces of equipment. The dust and chemicals released during the process not only pollute the air, but also deteriorate the health of the workers and people living in the nearby areas.

Domestic Sources: The household cleaning products and paints contain toxic chemicals that are released in the air. The smell from the newly painted walls is the smell of the chemicals present in the paints. It not only pollutes the air but also affects breathing.

Effects of Air pollution:

- **Diseases:** Air pollution has resulted in several respiratory disorders and heart diseases among humans. The cases of lung cancer have increased in the last few decades. Children

living near polluted areas are more prone to pneumonia and asthma. Many people die every year due to the direct or indirect effects of air pollution.

- **Global Warming:** Due to the emission of greenhouse gases, there is an imbalance in the gaseous composition of the air. This has led to an increase in the temperature of the earth. This increase in earth's temperature is known as global warming. This has resulted in the melting of glaciers and an increase in sea levels. Many areas are submerged underwater.
- **Acid Rain:** The burning of fossil fuels releases harmful gases such as nitrogen oxides and sulphur oxides in the air. The water droplets combine with these pollutants, become acidic and fall as acid rain which damages human, animal and plant life.
- **Ozone Layer Depletion:** The release of chlorofluorocarbons, and hydrochlorofluorocarbons in the atmosphere is the major cause of depletion of the ozone layer. The depleting ozone layer does not prevent the harmful ultraviolet rays coming from the sun and causes skin diseases and eye problems among individuals.
- **Effect on Animals:** The air pollutants suspend in the water bodies and affect aquatic life. Pollution also compels the animals to leave their habitat and shift to a new place. This renders them stray and has also led to the extinction of a large number of animal species.

Prevention and control of Air Pollution:

• Inputs that do not contain the pollutants.

• Operating process to minimize generation of the pollutants.

• Replacing the process with one does not generate the pollutant.

• Removing the pollutants from the process.

• Substitution of raw materials.

Global Warming

• Troposphere, the lowermost layer of the atmosphere, traps heat by a natural process due to the presence of certain gases. This effect is called Green House Effect as it is similar to the warming effect observed in the horticultural greenhouse made of glass.

• The amount of heat trapped in the atmosphere depends mostly on the concentrations of "heat trapping" or "greenhouse" gases and the length of time they stay in the atmosphere. The major greenhouse gases are carbon dioxide, ozone, methane, nitrous oxide, chlorofluorocarbons (CFCs) and water vapours.

• Heat trapped by greenhouse gases in the atmosphere keeps the planet warm is called global warming.

Greenhouse Effect

The enhanced greenhouse effect will not only cause global warming but will also affect various other climatic and natural processes.

(i) Global temperature increase: It is estimated that the earth's mean temperature will rise between 1.5 to 5.5°C by 2050 if input of greenhouse gases continues to rise at the present rate. Even at the lower value, earth would be warmer than it has been for 10,000 years.

(ii) Rise in Sea Level: With the increase in global temperature sea water will expand. Heating will melt the polar ice sheets and glaciers resulting in further rise in sea level. Current models indicate that an increase in the average atmospheric temperature of 3°C would raise the average global sea level by 0.2–1.5 meters over the next 50–100 years.

(iii) Effects on Human Health: The global warming will lead to changes in the rainfall pattern in many areas, thereby affecting the distribution of vector-borne diseases like malaria, filariasis, elephantiasis etc.

Warmer temperature and more water stagnation would favour the breeding of mosquitoes, snails and some insects, which are the vectors of such diseases. Higher temperature and humidity will increase/aggravate respiratory and skin diseases.

Ozone Layer Depletion

For the last 450 million years the earth has had a natural sunscreen in the stratosphere called the ozone layer. This layer filters out harmful ultraviolet radiations from the sunlight and thus protects various life forms on the earth. Ozone is a form of oxygen. The molecule of oxygen contains two atoms whereas that of ozone contains three (O3). In the stratosphere ozone is continuously being created by the absorption of short wave-length ultraviolet (UV) radiations. Ultraviolet radiations less than 242 nanometres decompose molecular oxygen into atomic oxygen (O) by photolytic decomposition.

$O_2 + hv \rightarrow O + O$

The atomic oxygen rapidly reacts with molecular oxygen to form ozone.

$O + O_2 + M \rightarrow O_3 + M$

(M is a third body necessary to carry away the energy released in the reaction).

Ozone thus formed distributes itself in the stratosphere and ab-sorbs harmful ultraviolet radiations (200 to 320 nm) and is continuously being converted back to molecular oxygen.

$O_3 + hv \rightarrow O_2 + O$

Absorption of UV radiations results in heating of the stratosphere. The net result of the above reactions is an equilibrium concentration of ozone. Ozone concentration in about 24 km of the

stratosphere i.e. from 16 km to 40 Km away from earth is about 10 ppm (as com-pared to 0.05 ppm concentration of harmful tropospheric ozone). This equilibrium is disturbed by reactive atoms of chlorine, bromine etc. which destroy ozone molecules and result is thinning of ozone layer generally called ozone hole. The amount of atmospheric ozone is measured by 'Dobson Spectrometer' and is expressed in Dobson units (DU). One DU is equivalent to a 0.01 mm thickness of pure ozone at the density it would possess if it were brought to ground level (1atm) pressure. Normally over temperate latitude its concentration is about 350 DU, over tropics it is 250 DU whereas at sub polar regions (except when ozone thinning occurs) it is on an average 450 DU. It is because of the stratospheric winds which transport ozone from tropical towards polar regions.

WATER POLLUTION

The earth has more than two-thirds of its surface covered with water, distributed in the oceans, rivers, lakes and streams. That is a lot of water; however, less than 0.3% is accessible for human consumption. Water pollution can be defined as the contamination of water bodies. Water pollution is caused when water bodies such as rivers, lakes, oceans, groundwater and aquifers get contaminated with industrial and agricultural effluents. When water gets polluted, it adversely affects all life forms that directly or indirectly depend on this source. The effects of water contamination can be felt for years to come.

Causes of water pollution:

1. Industrial Waste: Industries and industrial sites across the world are a major contributor to water pollution. Many industrial sites produce waste in the form of toxic chemicals and pollutants, and though regulated, some still do not have proper waste management systems in place. Industrial waste from agricultural sites, mines and manufacturing plants can make its way into rivers, streams and other bodies of water that lead directly to the sea. The toxic chemicals in the waste produced by these industries not only have the potential to make water unsafe for human consumption, they can also cause the temperature in freshwater systems to change, making them dangerous for many water dwelling organisms.

2. Marine Dumping: The process of marine dumping is exactly what it sounds like, dumping garbage into the waters of the ocean. It might seem crazy, but household garbage is still collected and dumped into oceans by many countries across the world. Most of these items can take anywhere from two to 200 years to decompose completely.

3. Sewage and Wastewater: Harmful chemicals, bacteria and pathogens can be found in sewage and wastewater even when it's been treated. Sewage and wastewater from each household is released into the sea with fresh water. The pathogens and bacteria found in that wastewater breed disease, and therefore are a cause of health-related issues in humans and animals alike.

4. Oil Leaks and Spills: The age-old phrase “like water and oil” is used when describing two things that do not mix easily or at all. Just as the saying states, water and oil do not mix, and oil does not dissolve in water. Large oil spills and oil leaks, while often accidental, are a major cause of water pollution. Leaks and spills often are caused by oil drilling operations in the ocean or ships that transport oil.

5. Agriculture: In order to protect their crops from bacteria and insects, farmers often use chemicals and pesticides. When these substances seep into the groundwater, they can harm animals, plants and humans. Additionally, when it rains, the chemicals mix with rainwater, which then flows into rivers and streams that filter into the ocean, causing further water pollution.

6. Global Warming: Rising temperatures due to global warming are a major concern in terms of water pollution. Global warming causes water temperatures to rise, which can kill water-dwelling animals. When large die-offs occur, it further pollutes the water supply, exacerbating the issue.

7. Radioactive Waste: Radioactive waste from facilities that create nuclear energy can be extremely hazardous to the environment and must be disposed of properly. This is because uranium, the element used in the creation of nuclear energy, is a highly toxic chemical.

Effects of Water Pollution:

- Water bodies in the vicinity of urban areas are extremely polluted. This is the result of dumping garbage and toxic chemicals by industrial and commercial establishments.
- Water pollution drastically affects aquatic life. It affects their metabolism, and behaviour, and causes illness and eventual death. Dioxin is a chemical that causes a lot of problems from reproduction to uncontrolled cell growth or cancer. This chemical is bioaccumulated in fish, chicken and meat. Chemicals such as this travel up the food chain before entering the human body.
- The effect of water pollution can have a huge impact on the food chain. It disrupts the food chain. Cadmium and lead are some toxic substances, these pollutants upon entering the food chain through animals (fish when consumed by animals, humans) can continue to disrupt at higher levels.
- Humans are affected by pollution and can contract diseases such as hepatitis through faecal matter in water sources. Poor drinking water treatment and unfit water can always cause an outbreak of infectious diseases such as cholera, etc.
- The ecosystem can be critically affected, modified and destructured because of water pollution.

Control measures for water pollution:

Water pollution, to a larger extent, can be controlled by a variety of methods. Rather than releasing sewage waste into water bodies, it is better to treat them before discharge. Practising this can reduce the initial toxicity and the remaining substances can be degraded and rendered harmless by the water body itself. If the secondary treatment of water has been carried out, then this can be reused in sanitary systems and agricultural fields.

A very special plant, the Water Hyacinth can absorb dissolved toxic chemicals such as cadmium and other such elements. Establishing these in regions prone to such kinds of pollutants will reduce the adverse effects to a large extent.

Some chemical methods that help in the control of water pollution are precipitation, the ion exchange process, reverse osmosis, and coagulation. As an individual, reusing, reducing, and recycling wherever, possible will advance a long way in overcoming the effects of water pollution.

SOIL POLLUTION

Soil pollution refers to anything that causes contamination of soil and degrades the soil quality. It occurs when the pollutants causing the pollution reduce the quality of the soil and convert the soil inhabitable for microorganisms and macro organisms living in the soil.

Some of the most hazardous soil pollutants are xenobiotics – substances that are not naturally found in nature and are synthesized by human beings. The term 'xenobiotic' has Greek roots – 'Xenos' (foreigner), and 'Bios' (life). Several xenobiotics are known to be carcinogens.

Causes of soil pollution:

- Industrial Pollution: The discharge of industrial waste into soils can result in soil pollution. In India, as mining and manufacturing activities are increasing rapidly, soil degradation is also increasing. The extraction of minerals from the earth is responsible for affecting soil fertility. Whether it is iron ore or coal, the by-products are contaminated, and they are disposed of in a manner that is not considered safe. As a result, the industrial waste stays on the soil surface for a long duration and makes it unsuitable for further use.
- Agricultural Activities: The use of insecticides and pesticides for a long period can cause soil pollution. Repetitive use can cause insects and pests to become resistant to it. Instead of killing pests and insects, it degrades the soil quality. They are full of chemicals that are not produced in nature and cannot be broken down by them. As a result, they seep into the ground after they mix with water and slowly reduce the fertility of the soil. Plants absorb many of these pesticides, and after decomposition cause soil pollution.
- Waste Disposal: Disposal of plastics and other solid waste is a serious issue that causes soil pollution, disposal of electrical items such as batteries causes an adverse effect on the

soil due to the presence of harmful chemicals. Eg: lithium present in batteries can cause the leaching of soil. Human waste such as urine, faeces, diapers, etc. is dumped directly in the land. It causes both soil and water pollution.

- Acid Rain: It is caused when pollutants present in the air mix with the rain and fall back on the ground. The polluted water could dissolve away some of the essential nutrients found in soil and change the structure of the soil thus making it unsuitable for agriculture.
- Heavy Metals: The presence of heavy metals (such as lead and mercury) in very high concentrations present in soils can cause them to become highly toxic for human beings.
- Nuclear Waste: It can also lead to soil degradation.
- Oil Spills: Oil leaks can happen during the storage or transport of chemicals, and the chemicals present in the fuel deteriorates the quality of soil and make them unsuitable for further cultivation, chemicals can also enter into the groundwater through the soil, and hence it will make water undrinkable.

Effects of Soil Pollution:

- Effects of Soil Pollution on Human Health: The contamination of soil has a major consequence on human health. Crops and plants that are grown on polluted soil absorb most of the pollution and then pass them to humans. Living, working, or playing in contaminated soil can lead to respiratory diseases, skin diseases, and other health problems. Diseases caused by soil pollution include Irritation of the skin and the eyes, Headaches, nausea, vomiting, coughing, pain in the chest, and wheezing.
- Effects on Plants: In such a short period of time, plants are unable to adapt to the soil change chemistry. Fungi and bacteria found in the soil that bind them together start to decline, which creates an additional problem in soil erosion. Regular use of chemical fertilizers, inorganic fertilizers, pesticides will decrease the fertility of the soil and alter the structure of soil. This will lead to a decrease in soil quality and poor quality of crops. The fertility of the soil diminishes slowly, making land unsuitable for agriculture and any local vegetation to survive.
- Effects on the Ecosystem: The soil is an important habitat for different types of microorganisms, birds, and insects. Thus, change in the chemistry of soil can negatively impact the lives of living organisms and can result in the gradual death of many organisms.

Control and prevention to Soil Pollution:

Soil pollution is a complex issue that must be addressed. It is important that we all understand the importance of soil to our survival. The earlier we recognize the problem, the simpler it will be to solve the problem of soil pollution. It's a complicated problem that requires everyone's participation, from individuals to the government. A few methods for reducing soil pollution are listed below.

- Reduced Use of Chemical Fertilizers: Chemical fertilizers are more damaging than helpful. While the right quantity can help the soil become more fertile, too much might potentially poison it. Chemical fertilizers in excess could harm the soil in a variety of ways. It has the ability to affect the soil's pH values.
- Reforestation and Afforestation Should be promoted: Soil erosion, which is produced by deforestation, is one of the major sources of soil pollution. With an ever-increasing population, it is only logical that mankind requires more and more room to expand their civilization. It is frequently accomplished at the expense of soil health. Reforestation of a deforested area should be encouraged to prevent this from happening.
- Recycle and Reuse Products: These measures not only reduce waste output, but they also reduce soil pollution. Plastic now makes up a significant portion of the waste flow. The great majority of these wastes are buried in landfills.
- Promote Use of Natural Manure: One of the best sources of nutrients for the soil is natural manure. It's 100% natural and safe. It restores the soil's critical nutrients and improves its overall health. It doesn't produce any toxic by-products that could harm the soil or the environment.

MARINE POLLUTION

Marine pollution is a combination of chemicals and trash, most of which comes from land sources and is washed or blown into the ocean. This pollution results in damage to the environment, to the health of all organisms, and to economic structures worldwide.

Causes of Marine Pollution:

- **Nonpoint source pollution (runoff):** Nonpoint source pollution is the accumulation of pollution from small sources that can’t be exactly pinpointed. Examples include the pollution created by individual cars, boats, farms and construction sites. Nonpoint source pollution typically becomes ocean pollution via runoff, which occurs when rain or snow moves pollutants from the ground into the ocean. For instance, after a heavy rainstorm, water flows off roads into the ocean, taking oil left on streets from cars with it. But wind can transfer dirt and other debris from nonpoint sources and deposit these pollutants on the ocean’s surface.
- **Intentional discharge:** Manufacturing plants in many areas of the world release toxic waste into the ocean, including mercury. While it has been banned in the USA, sewage also contributes to ocean pollution. Meanwhile, plastic waste poses a particularly tough challenge; according to the Ocean Conservancy, eight million metric tons of plastic go into our oceans every year.
- **Oil spills:** Ships are major contributors to ocean pollution, especially when crude oil spills occur. Crude oil lasts for years in the ocean and is difficult to clean up. You can find a history of oil spills on the NOAA site, but the largest one in recent history was, by

far, the 2010 Deepwater Horizon well blowout in the Gulf of Mexico, which spilled roughly 134 million gallons of oil into the ocean.

- **Littering:** Atmospheric pollution — a type of ocean pollution where objects are carried by the wind to the ocean — is often caused by littering. It includes single-use plastics (such as plastic bags) and styrofoam containers which can take hundreds of years to biodegrade. It is estimated that roughly 1 trillion plastic bags are used worldwide per year.
- **Ocean mining:** Deep-sea ocean mining causes ocean pollution and ecosystem disruption at the lowest levels of the ocean. Drilling for substances such as cobalt, zinc, silver, gold and copper creates harmful sulfide deposits deep in the ocean.

Effects of Marine Pollution:

Ocean pollution has many consequences that directly and indirectly affect marine life, as well as humans. Here are some of the most common effects of ocean pollution:

- Harmful to marine species: Sea animals are frequent victims of ocean pollution. Oil spills, for instance, will ensnare and suffocate marine animals by permeating their gills. When the oil gets into seabird feathers, they may not be able to fly or feed their young. Animals that aren't killed by crude oil may suffer from cancer, behavioral changes and become unable to reproduce. Marine animals also mistake small plastic debris for food or become entangled in or strangled by plastic bags and discarded fishing nets. Animals most vulnerable to harm from plastic debris in the ocean include dolphins, fish, sharks, turtles, seabirds and crabs.
- Depletion of oxygen in seawater: As excess debris in the ocean slowly degrades over many years, it uses oxygen to do so, resulting in less oxygen in the ocean. Low levels of oxygen in the ocean lead to the death of ocean animals such as penguins, dolphins, whales and sharks. Excess nitrogen and phosphorus in seawater also cause oxygen depletion. When a great deal of oxygen depletion occurs in an area of the ocean, it can become a dead zone (see below) where very little marine life can survive.
- A threat to human health: Pollutants in the ocean make their way back to humans. Small organisms ingest toxins and are eaten by larger predators, many of which are seafood that we eventually eat. When the toxins in contaminated animals get deposited in human tissue, it can lead to long-term health conditions, cancer and birth defects. For example, phytoplankton will absorb methylmercury (bacteria-converted mercury carried down to the ocean's surface). This makes its way up the food chain when zooplankton eat the phytoplankton, then small fish eat the zooplankton, then bigger fish eat the smaller fish. By the time you reach a fish the size of a swordfish, they can contain a very high mercury load (in fact, swordfish contain one of the highest mercury loads).

Control and prevention of Marine Pollution:

Solutions for marine pollution include prevention and cleanup. Disposable and single-use plastic is abundantly used in today's society, from shopping bags to shipping packaging to plastic bottles. Changing society's approach to plastic use will be a long and economically challenging process. Cleanup, in contrast, may be impossible for some items. Many types of debris (including some plastics) do not float, so they are lost deep in the ocean. Plastics that do float tend to collect in large "patches" in ocean gyres. The Pacific Garbage Patch is one example of such a collection, with plastics and microplastics floating on and below the surface of swirling ocean currents between California and Hawaii in an area of about 1.6 million square kilometers (617,763 square miles), although its size is not fixed. These patches are less like islands of trash and, as the National Oceanic and Atmospheric Administration says, more like flecks of microplastic pepper swirling around an ocean soup. Even some promising solutions are inadequate for combating marine pollution. So-called "biodegradable" plastics often break down only at temperatures higher than will ever be reached in the ocean.

- **Reduce chemical fertilizer use:** Excess chemical fertilizer eventually makes its way into the oceans. Choose organic fertilizers, which tend to be lower in nutrients, and use them at half strength or half as often as suggested.
- **Opt for reusable bottles and utensils:** Single-use plastic bottles, eating utensils and straws, are massive ocean polluters. Rather than contributing to the threat to marine life, opt for reusable bottles and utensils. When using reusable bottles, BPA-Free or glass bottles are best. Reusable straws made from silicone or metal are great options to use as well. They get the job done, can be used multiple times and stay out of our landfills and oceans.
- **Properly dispose of plastics and other trash:** One of the simplest ways to reduce ocean pollution is to properly dispose of plastics and other recyclable materials, so they don't end up in the ocean. In outdoor spaces, such as beaches and parks, dispose of trash in a secure receptacle or take it home with you.

NOISE POLLUTION

The word noise is derived from the Latin word 'Nausea', which means sickness in which one feels the need to vomit. Noise is the unpleasant and undesirable sound which leads to discomfort in human beings. The intensity of sound is measured in decibels (dB). The faintest sound that the human ear can hear is 1 Db. Due to increasing noise around the civilizations; noise pollution has become a matter of concern. Some of its major causes are vehicles, aircraft, industrial machines, loudspeakers, crackers, etc. When used at high volume, some other appliances also contribute to noise pollution, like television, transistor, radio, etc.

Causes of Noise Pollution:

Following are the causes and sources of noise pollution:

- Industrialization: Industrialization has led to an increase in noise pollution as the use of heavy machinery such as generators, mills, huge exhaust fans are used, resulting in the production of unwanted noise.
- Vehicles: Increased number of vehicles on the roads is the second reason for noise pollution.
- Events: Weddings, public gatherings involve loudspeakers to play music resulting in the production of unwanted noise in the neighborhood.
- Construction sites: Mining, construction of buildings, etc. add to the noise pollution.

Following are the examples of noise pollution:

- Unnecessary usage of horns
- Using loudspeakers either for religious functions or for political purposes
- Unnecessary usage of fireworks
- Industrial noise
- Construction noise
- Noise from transportation such as railway and aircraft

Effects of Noise Pollution:

Noise pollution can be hazardous to human health in the following ways:

- Hypertension: It is a direct result of noise pollution which is caused due to elevated blood levels for a longer duration.
- Hearing loss: Constant exposure of human ears to loud noise that are beyond the range of sound that human ears can withstand damages the eardrums, resulting in loss of hearing.
- Sleeping disorders: Lack of sleep might result in fatigue and low energy level throughout the day affecting everyday activities. Noise pollution hampers the sleep cycles leading to irritation and an uncomfortable state of mind.
- Cardiovascular issues: Heart-related problems such as blood pressure level, stress and cardiovascular diseases might come up in a normal person and a person suffering from any of these diseases might feel a sudden shoot up in the level.

Control and prevention of Noise Pollution:

- Honking in public places like teaching institutes, hospitals, etc. should be banned.
- In commercial, hospital, and industrial buildings, adequate soundproof systems should be installed.
- Musical instruments’ sound should be controlled to desirable limits.

- Dense tree cover is useful in noise pollution prevention.
- Explosives should not be used in forest, mountainous and mining areas**.**

THERMAL POLLUTION

Thermal pollution is a rapid change in temperature in a natural body of water. This pollution is most often caused by heated discharge from an industrial facility or another human activity. Thermal pollution can result in disruptions in natural systems and stress, disease, or even death for affected organisms.

Causes of Thermal Pollution:

Natural phenomena such as wildfires, volcanoes, and underwater thermal vents can cause thermal pollution. However, it is more often the result of an industrial process or facility using large amounts of water from a natural source and releasing heated wastewater.

- **Power Plants and Industrial Facilities:** Thermoelectric power plants fueled by coal, natural gas, nuclear, or biomass and other waste products constitute significant causes of thermal pollution. Power plants are typically built next to a river, lake, or ocean, which provide a steady supply of water. This gets converted to steam that drives turbines to generate electricity. Water is also used to cool machinery, which becomes very hot. The water absorbs heat, and what doesn't evaporate is typically discharged back to its source. In addition to power plants, other industrial facilities—such as petroleum refineries, pulp and paper mills, chemical plants, and steel mills—contribute to thermal pollution.
- **Desalination Plants:** Desalination plants also utilize once-through cooling. More than half of the seawater used in desalination gets flushed back into the ocean as wastewater, often at an elevated temperature. In some parts of the world, desalination plants are clustered together, pouring massive amounts of heated, briny wastewater into shallow coastal regions. This can raise seawater temperature and salinity significantly.
- **Wastewater, Erosion, and Deforestation:** Not all wastewater gets treated before it is released into water bodies. Untreated sewage, and agricultural runoff can create thermal pollution in nearby water sources as the runoff is often warmer than the streams, lakes, or ocean they flow into. Human land-use changes cause thermal pollution as well. Deforestation for timber harvesting or clearing land for crops and livestock grazing provokes erosion alongside rivers and streams, which leads to wider, shallower stream beds more prone to warming. In addition, clearing trees and vegetation from lakeshores and riverbanks creates more sun exposure, provoking water warming.
- **Natural Causes:** Natural causes like volcanoes, geothermal vents, and hot springs under the oceans and seas can trigger warm lava to raise the temperature of water bodies. Lightening can also introduce a massive amount of heat into the oceans. This means that the water source’s overall temperature will rise, significantly impacting the environment.

Effects of Thermal Pollution:

- **Decrease in DO (Dissolved Oxygen) Levels:** The warm temperature reduces the levels of DO (Dissolved Oxygen) in the water. Warm water holds relatively less oxygen than cold water. The decrease in DO can suffocate plants and animals such as fish, amphibians, and copepods, which may give rise to anaerobic conditions. In addition, warmer water allows algae to flourish on the water's surface, and that can decrease oxygen levels in the water over the long term.
- **Increase in Toxins:** With the constant flow of high-temperature discharge from industries, a huge increase in toxins regurgitated into the natural body of water. These toxins may contain chemicals or radiation that may have a harsh impact on the local ecology and make them susceptible to various diseases.
- **Loss of Biodiversity:** A dent in the biological activity in the water may cause a significant loss of biodiversity. Changes in the environment may trigger certain species of organisms to shift their base to some other place, while a significant number of species may shift in because of warmer waters.
- **Ecological Impact:** A sudden thermal shock can result in the mass killing of fish, insects, plants, or amphibians. Hotter water may prove favorable for some species, while it could be lethal for others. Many aquatic species are sensitive to small temperature changes, such as one degree Celsius that can cause significant changes in organism metabolism and other adverse cellular biology effects.
- **Affects Reproductive Systems:** A significant halt in the reproduction of marine wildlife (although this may be true, reproduction can still occur between fish – but the likelihood of defects in newborns is significantly higher) can happen due to increasing temperatures, as reproduction can happen within a certain range of temperatures. Excessive temperatures can cause the release of immature eggs or can prevent the normal development of certain eggs.
- **Increases Metabolic Rate:** Thermal pollution increases the metabolic rate of organisms. High temperatures increase enzyme activity, causing organisms to consume more food than what is normally required if their environment were not changed. It disrupts the stability of the food chain and alters the balance of species composition.
- **Migration:** The warm water can also cause particular species of organisms to migrate to a suitable environment that would cater to their requirements for survival. This can result in a loss for those species that depend on them for their daily food as their food chain is interrupted.

Control & Prevention to Thermal Pollution:

- **Cooling Ponds:** Cooling ponds or reservoirs are the simplest methods of controlling thermal discharges. In cooling ponds, heated effluents on the surface of water maximize heat dissipation into the atmosphere and minimize the area and volume of water. This is

the simplest and cheapest method that cools the water to a considerably low temperature. However, the method alone is less desirable and inefficient in terms of air-water contact.

- **Cooling Towers:** After using water from water sources for cooling purposes, it is returned to the water body after passing through the condenser, which is termed the cooling process. Therefore, cooling towers are designed to control water temperature to make the cooling process more effective. Cooling towers are mainly used to dissipate the recovered waste heat to eliminate thermal pollution problems.
- **Artificial Lake:** Artificial lakes are man-made water bodies that offer a possible alternative. The heated effluents may be discharged into the lake at one end, and the water may be withdrawn from the other end for cooling purposes. The heat is eventually dissipated through evaporation. However, these lakes have to be rejuvenated continuously.
- **Water Recycling:** Industrially treated water can be recycled for domestic use or industrial heating so that the problem of thermal pollution can be mitigated.
- **Other Applications:** The thermal discharge (rejected heat) of power plants can be used for other purposes, like:
    - Industrial and space heating.
    - Biological applications such as soil warming.
    - Fish culture, livestock shelters, and heating greenhouses.
    - Most of these potential physical applications are applicable in colder regions or locations.

Above all else, the most important thing to consider is that the effects of thermal pollution greatly outweigh the human need for it to be done. Plants and industries have found successful ways around thermal pollution, but many of them are not practicing it because it's simply easier to work from the traditional model.

NUCLEAR HAZARDS

Nuclear hazards and incidents generally refer to incidents involving (1) the release of significant levels of radioactive materials or (2) exposure of workers or the general public to radiation. The primary concerns following a nuclear incident or accident is the public health impact from direct exposure to a radioactive plume, inhalation of radioactive materials, ingestion of contaminated food, water and milk, and long term exposure to deposited radioactive materials in the environment that may lead to either acute (radiation sickness or death) or chronic (cancer) health effects.

Causes:

Nuclear power production almost always causes nuclear pollution. Of course, nuclear plants try to limit the number of radioactive contaminants they release. But, radioactive material and nuclear waste still find a way to enter the environment.

- **Nuclear Waste Disposal:** Nuclear plants use fuel to run. When the plant cannot use that fuel any longer, it must dispose of the spent fuel. Most nuclear fuels have a half-life of up to four billion years. It means that the energy can remain radioactive for up to four billion years! When we humans first discovered and started using nuclear energy, a lot of the nuclear waste would end up in the oceans. Even today, we continue to dump our nuclear waste in the seas. The Pacific Ocean is a famous nuclear waste dumping ground for many countries.

  Apart from dumping it in oceans, some nuclear plants store their spent fuel in underground pools. Nuclear fuel is hot and needs to cool before it can be disposed of. However, storing nuclear waste underground puts groundwater and the surrounding land at risk of contamination. If the surrounding area is cropland, radioactive materials can enter the crops and, ultimately, our food chain.
- **Nuclear Accidents:** We find some of the most concentrated areas of nuclear pollution in the region surrounding accidents at nuclear power plants. History has witnessed only a handful of such events. But the effects are catastrophic and prevail many years after the accident.

  Many will remember the Chornobyl nuclear disaster of 1986 as one of the most infamous and devastating nuclear accidents. The disaster occurred in modern-day Ukraine. The failure of a nuclear reactor destroyed the entire nuclear plant. Eastern Europe witnessed air pollution caused by the release of hazardous, radioactive materials. It affected thousands of people. Many people died from exposure to radioactivity from the Chornobyl reactor.

- **Nuclear Weapons:** The Second World War saw the extensive use of nuclear weapons by countries. We will all recall when our history professor explained how the Japanese cities of Hiroshima and Nagasaki were destroyed by atomic bombs. Since the Second World War, countries have been racing to develop nuclear weapons in the name of defense. Countries test their nuclear weapons by firing them into the atmosphere. The explosion in the atmosphere returns back debris to the Earth as radiation. When this radiation settles on vegetation and in our seas and oceans, it enters the food chain.

Effects of Nuclear hazards:

- The radiation from nuclear pollution has enough energy to damage living cells and their DNA. The cells in our body are capable of repairing this damage. However, if our bodies fail to repair the damage correctly, cells may die or eventually turn cancerous.
- Being exposed to extremely high levels of radiation can result in skin burns and acute radiation syndrome (also known as radiation sickness). High radiation levels can also cause long-term health effects like cancer and cardiovascular disease.

- Exposure to low radiation levels does not result in immediate health effects. However, it is a minor contributor to our overall cancer risk.
- Exposure to a high level of radiation within a short time span causes symptoms such as nausea and vomiting. These symptoms may appear within hours of exposure. But it may result in death over the following days or weeks. This is what scientists call acute radiation syndrome or radiation sickness.
- The level of radiation required to develop acute radiation syndrome is equivalent to getting 18,000 chest X-rays within a few minutes. Acute radiation syndrome is extremely rare. Scientists mainly observe the syndrome in people exposed to a nuclear explosion or in the vicinity of a highly radioactive source rupture.
- Studies conducted on numerous atomic bomb survivors and radiation industry workers have shown that radiation exposure increases the chance of getting cancer. The higher the dose of exposure, the greater the risk of developing cancer.

Control measures:

- **Containment of Nuclear Waste:** Nuclear radiation is a form of heat transfer. While radiation can occur in almost any condition, heat increases the amount of radiation. More radiation means a higher health risk. Scientists recently found that apart from the ash released from nuclear plants, even coal ash, and wood ash contain radiation because of their heat. Therefore, we must store nuclear waste in cool places, away from a heat source.
- **Law Enforcement:** We need laws that protect human health and the environment from nuclear and radioactive radiation. Federal agencies in every country must establish radiation exposure standards and limits. National governments must also develop standards for nuclear power plants. They must implement strict actions against nuclear plants for failing to comply with environmental and health regulations.
- **Individual Prevention Measures:** You should regularly test your home for radon. The internet can find you many consulting services and inexpensive testing kits.